

Course Code: CS1107		
Name of the Programme: BCA (Hons.)		
Name of the Course: Digital Logic		
Semester: I		
Course Credit	No. of Hours Per Week (L+ T + P)	Total No. of Teaching Hours
3	1:1:2	60
Course Objectives:		
a) Introduce the fundamental principles of digital logic design and its role in computing systems.		
b) Develop the ability to analyze and design basic digital circuits using combinational and sequential logic.		
c) Explore the structure, function, and interaction of core computer architecture components.		
d) Apply theoretical knowledge to practical scenarios, laying a strong foundation for advanced studies in computer systems and engineering.		

Course outline (Syllabus of the course)

Module - 1 (Digital Systems and Binary Numbers, Boolean Algebra and Logic Gates)	No. of Hours
	3 Hrs.
Digital systems, binary numbers, number base conversions, octal and hexadecimal numbers, complements, signed binary numbers, binary codes, error detection and error correction codes.	
Basic definitions, axiomatic definition of Boolean algebra, basic theorems and properties of Boolean algebra, Boolean functions, canonical and standard forms.	
Module - 2 (Gate Level Minimization)	No. of Hours
	3 Hrs.
The k-map method, four-variable map, five-variable map, product of sums simplification, don't-care conditions, NAND and NOR implementation	
Module - 3 (Combinational Circuits)	No. of Hours
	3 Hrs.

Design procedure, Binary Adder, Binary Subtractor, Decoders, Encoders, Multiplexers, and Demultiplexers.	
Module - 4 (Synchronous Sequential Logic, Registers and Counters)	No. of Hours
	3 Hrs.
Sequential circuits, latches, flip-flops Registers, shift registers, ripple counters	
Module - 5 (Memory and Programmable Logic)	No. of Hours
	3 Hrs.
Introduction, Random access memory, memory decoding, read only memory, programmable logic array, programmable array logic, sequential programmable devices.	

Course Outcomes:
a) Understand binary number systems, codes, and error detection/correction techniques in digital systems.
b) Apply Boolean algebra to simplify logic expressions and design circuits using logic gates and standard forms.
c) Analyze and design combinational and sequential circuits such as adders, multiplexers, and counters for practical applications.
d) Evaluate the role and functionality of memory and programmable logic devices like RAM, ROM, and PLDs in digital systems.

Textbooks	
1	Mano, M. M., & Ciletti, M. D. (2017). Digital Design (6th ed.). Pearson. ISBN: 978-0134549897
2	Roth, C. H., Kinney, L. L., & John, C. H. (2016). Fundamentals of Logic Design (7th ed.). Cengage Learning. ISBN: 978-1305635142

Reference Books	
1	Rao, C. V. S. (2009). <i>Switching and logic design</i> (3rd ed.). Pearson Education. ISBN: 9978-81-317-0183-6
2	Wakerly, J. F. (2018). <i>Digital Design: Principles and Practices</i> (5th ed.). Pearson. ISBN: 978-0134460093
3	Harris, D. M., & Harris, S. L. (2021). <i>Digital Design and Computer Architecture</i> (3rd ed.). Morgan Kaufmann. ISBN: 978-0128200650

Reference Links	
1	https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/
2	https://www.javatpoint.com/digital-electronics
3	https://www.tutorialspoint.com/digital-electronics/index.htm

Tutorial Component	
1.	Perform number base conversions between binary, octal, decimal, and hexadecimal systems.
2.	Explain the concept of complements (1's and 2's) and perform arithmetic operations using signed binary numbers.
3.	Encode given data using binary codes and design an error detection/correction mechanism using Hamming codes.
4.	Apply the axiomatic definition of Boolean algebra to prove basic theorems and simplify Boolean expressions.
5.	Represent Boolean functions in canonical forms (SOP and POS) and standard forms.
6.	Simplify a given Boolean function using Karnaugh Maps (K-map) for up to five variables and implement the simplified expression.
7.	Design and implement a Boolean expression using NAND and NOR gates only.
8.	Design a combinational circuit for a binary adder and implement it using logic gates.
9.	Design a combinational circuit for a binary subtractor and explain its operation.
10.	Create encoder, decoder, multiplexer, and demultiplexer circuits and explain their applications in digital systems.

11.	Design a sequential circuit using flip-flops and explain the differences between latches and flip-flops.
12.	Design and implement shift registers for specific bit-length operations (e.g., serial-in serial-out, serial-in parallel-out).
13.	Construct a ripple counter circuit and analyze its timing diagram.
14.	Explore the structure of memory devices, explain RAM and ROM functionality, and design a memory decoding circuit.
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Laboratory Component	
1	Verification of truth tables of the following Logic gates Two input (i) OR (ii) AND (iii) NOR (iv) NAND (v) Exclusive-OR (vi) Exclusive-NOR
2	Design a simple combinational circuit with four variables and obtain minimal expression and verify the truth table using Digital Trainer Kit.
3	Verification of functional table of 3 to 8-line Decoder /De-Multiplexer.
4	4 variable logic function verification using 8 to1 multiplexer.
5	Design a full adder circuit and verify its functional table.
6	Verification of functional tables of (i) JK Edge triggered Flip–Flop (ii) JK Master Slave Flip–Flop (iii) D Flip-Flop
7	Design a four-bit ring counter using D Flip–Flops/JK Flip Flop and verify output
8	Design a four bit Johnson’s counter using D Flip-Flops/JK Flip Flops and verify output
9	Verify the operation of 4-bit Universal Shift Register for different Modes of operation.
10	Draw the circuit diagram of MOD-8 ripple counter and construct a circuit using T-Flip- Flops and Test It with a low frequency clock and sketch the output waveforms.



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11	Design MOD–8 synchronous counter using T Flip-Flop and verify the result and sketch the output waveforms.
12	(a) Draw the circuit diagram of a single bit comparator and test the output (b) Construct 7 Segment Display Circuit Using Decoder and 7 Segment LED and test it.

Lesson plan (template – 2)

Name of the Programme:	BCA	
Title of the Course:	Digital Logic	
Course code	CS1107	
Semester:	1	
Names of the Course Instructor(s)	Prof. Kaipangal Maneesh Rajan and Prof. Mohammed Danish	
Course credit	No. of hours per week (L + T + P)	Total no. of Teaching hours
3	1:1:2	60

Lesson Plan

Session	Module No.	Topic	Pedagogy / Activities	Pre-reading / Reference
1	1	Digital Systems, Binary Numbers	Lecture with examples	Textbook 1 Ch.1
2		Number base conversions, Complements	Tutorial + Practice problems	Textbook 1 Ch.1
3		Signed Binary Numbers, Binary Codes	LAB	Textbook 1 Ch.1
4		Error Detection and Correction Codes	LAB	Ref 3
5		Boolean Algebra Definitions	Lecture	Textbook 1 Ch.2
6		Theorems and Properties	Tutorial LAB	Textbook 1 Ch.2
7		Standard Forms, Boolean Functions	LAB + Problem solving session	Textbook 1 Ch.2
8		Other Logic Operations	Interactive board session (Lecture)	Ref 1
9		Digital Logic Gates	Demo + Hands-on practice (Tutorial)	Lab Manual
10	2	K-map Method - 2 variables	LAB demo	Textbook 1 Ch.3
11		K-map - 4 variable	LAB demo	Textbook 1 Ch.3



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12		Five-variable maps	Lecture + Practice	Ref 1
13		POS Simplification, Don't care	Tutorial + Q&A Lecture + Worksheet	Textbook 1 Ch.3
14		NAND/NOR Implementation	Lab session	Lab Manual
15		Prime Implicants Identification	Advanced session(LAB)	Textbook 1 Ch.3
16		Essential and Non-essential Implicants	Lecture+ Problem solving	Ref 2
17		Review & Practice	Quiz + discussion	Textbook 1 Ch.3
18		Activity - Boolean Simplification	Lab + Discussion	Lab Manual
19		Design Procedure	Lecture + Flowcharts	Textbook 1 Ch.4
20	3	Adders/Subtractors	Whiteboard Q&A(Tutorial)	Textbook 1 Ch.4
21		Binary Multiplier	Lecture + Lab	Textbook 1 Ch.4
22		Magnitude Comparator	Lab session	Lab Manual
23		Decoders/Encoders	Hands-on Lab	Textbook 1 Ch.4
24		Multiplexer/Demultiplexer	Demo + Practical	Ref 1
25		Verification Lab	Lab work	Lab Manual
26		Review & Applications	Discussion + Worksheet	Textbook 1
27		Mini Project - Combinational Circuit	Lab + Presentation	Lab Manual
28	4	Sequential Circuits & Flip-Flops	Lecture + Examples	Textbook 1 Ch.5
29		Latches, Clocking	Lab session	Lab Manual
30		State Reduction & Assignment	Problem Solving	Ref 2
31		Design Procedure for Sequential	Flowchart + Practice	Textbook 1
32		Registers and Types	Lecture + Chart	Textbook 1 Ch.5
33		Shift Registers	Hands-on Lab	Lab Manual
34		Ripple Counters	Lab + Discussion	Ref 3
35		Synchronous Counters	Implementation Lab	Lab Manual
36	5	Review of Sequential Circuits	Group Activity	Textbook 1
37		Introduction to Memory	Lecture + Q&A	Textbook 1 Ch.6
38		RAM and Organization	Demo + Worksheet	Textbook 1
39		Memory Decoding	Practical + Examples	Ref 2
40		Error Detection Techniques	Tutorial	Textbook 1



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41		Read Only Memory	Lab demonstration	Lab Manual
42		Programmable Logic Array	Hands-on Lab	Ref 1
43		Programmable Array Logic	Lab session	Lab Manual
44		Sequential Programmable Devices	Advanced Lecture	Textbook 1
45		Review & Assignment	Group Discussion	All Units
46	-	Lab Consolidation / Practice / Assessment	Hands-on + Viva	Lab Manual
47	-	Lab Consolidation / Practice / Assessment	Hands-on + Viva	Lab Manual
48	-	Lab Consolidation / Practice / Assessment	Hands-on + Viva	Lab Manual
49	-	Lab Consolidation / Practice / Assessment	Hands-on + Viva	Lab Manual
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60	-	Lab Consolidation / Practice / Assessment	Hands-on + Viva	Lab Manual

Assessment and Evaluation

Continuous Internal Evaluation (CIE)	
Components of CIE	Weightage of each component
- CIE 1 (MCQ + MSQ)	10 marks out of 20

- CIE 1 (Scenario Based)	10 marks out of 20
- CIE 2	25 marks out of 25
- CIE 3(LAB Execution)	15 marks out of 25
- CIE 3 (VIVA)	5 marks out of 25
- CIE 3 (Record)	5 marks out of 25

Semester End Exams (SEE)	
Mode of Exam	Pen and Paper
Weightage of exam	30%

Course Outcomes- Program Outcomes Matrix

Course Outcome	P01	P02	P03	P04	P05	P06	P07	P08
C01	3	1	2		2	-	-	-
C02	3	2	3		2	-	-	-
C03	3	2	3	1	2	-	1	-
C04	3	2	3	2	2	-	2	-
C05	2	1	2		3	-	2	-

Rubrics for evaluation of CIE (separate for each CIE component)

Programme Outcome	1. Complex Problem-Solving and Critical Thinking Graduates should be able to solve different kinds of problems in familiar and unfamiliar contexts, apply their learning to real-life situations. 2. Creative and Effective Communication Graduates should be able to demonstrate creativity by thinking in diverse ways, dealing with complex problems, innovating, and viewing situations from multiple perspectives. . 3. Analytical and Research Competence Graduates should be able to evaluate the reliability and relevance of evidence, identify logical flaws in arguments, and analyze and synthesize data from various sources to draw valid conclusions and support them with evidence. 4. Teamwork and Leadership Graduates should be able to work effectively and respectfully with diverse teams, facilitating cooperative efforts and acting together towards common goals. 5. Lifelong Learning and Digital Proficiency Graduates should demonstrate the ability to acquire new knowledge and skills for lifelong learning, adapt to changing workplace demands, work independently, and manage time and resources effectively. 6. Multicultural Competence and Ethical Values Graduates should demonstrate knowledge of multiple cultures, interact respectfully with diverse groups, lead diverse teams, and adopt a gender-neutral and empathetic approach. 7. Responsible Autonomy and Environmental Stewardship Graduates should demonstrate the ability to apply knowledge and skills independently, manage projects to completion, exercise responsibility, and ensure workplace safety and security. 8. Community Engagement and Empathy Graduates should demonstrate the capability to participate in community-engaged services and activities to promote societal well-being, and the ability to empathize by understanding the perspectives, experiences, and emotions of others.					
Total Marks	70					
CIE components	MCQ Evaluation (10 Marks)					
CIE-1 Graded Component 1 (Quiz) Weightage: 10	Marking Rule: ● Correct Answer: 2 marks ● Wrong Answer: 0 marks ● No partial marks					
CIE components	CO-PO					
CIE-1: Scenario Based	CO1	Rubrics Scenario Based Questions:				

Weightage: 10	PO1, PO2	<table><tr><td>Criteria</td><td>Excellent (7–10 marks)</td><td>Good (4–6 marks)</td><td colspan="2">Satisfactory (2–4 marks)</td></tr><tr><td>Understanding of Scenario</td><td>Demonstrates deep understanding; clearly identifies core issues and implications</td><td>Shows good understanding; identifies most key issues</td><td colspan="2">Basic understanding; some issues are identified, but lacks depth</td></tr><tr><td>Application of Concepts</td><td>Applies relevant concepts/theories accurately and insightfully</td><td>Applies most relevant concepts with minor errors</td><td colspan="2">Attempts to apply concepts, but with some misunderstandings or omissions</td></tr><tr><td>Problem Solving / Decision Making</td><td>Presents well-structured, feasible, and innovative solutions/decisions</td><td>Provides workable solutions with reasonable justification</td><td colspan="2">Solutions are simplistic or partially justified</td></tr><tr><td>Critical Thinking / Justification</td><td>Arguments are logical, well-structured, and critically analyzed with evidence/examples</td><td>Reasonable logic and structure; some critical analysis and justification</td><td colspan="2">Limited logic and critical thinking; basic justification provided</td></tr><tr><td>Clarity and Presentation</td><td>Clear, concise, well-organized, and free from major errors</td><td>Mostly clear and organized; minor errors</td><td colspan="2">Understandable but lacks clarity or organization</td></tr><tr><td colspan="5"></td></tr></table>				Criteria	Excellent (7–10 marks)	Good (4–6 marks)	Satisfactory (2–4 marks)		Understanding of Scenario	Demonstrates deep understanding; clearly identifies core issues and implications	Shows good understanding; identifies most key issues	Basic understanding; some issues are identified, but lacks depth		Application of Concepts	Applies relevant concepts/theories accurately and insightfully	Applies most relevant concepts with minor errors	Attempts to apply concepts, but with some misunderstandings or omissions		Problem Solving / Decision Making	Presents well-structured, feasible, and innovative solutions/decisions	Provides workable solutions with reasonable justification	Solutions are simplistic or partially justified		Critical Thinking / Justification	Arguments are logical, well-structured, and critically analyzed with evidence/examples	Reasonable logic and structure; some critical analysis and justification	Limited logic and critical thinking; basic justification provided		Clarity and Presentation	Clear, concise, well-organized, and free from major errors	Mostly clear and organized; minor errors	Understandable but lacks clarity or organization						
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CIE components	CO-PO	Continuous Internal Evaluation-2 (25 Marks)																																						
CIE-2 Mid Sem Examination (Theory) Weightage: 25	CO1, CO2, CO3 PO1, PO2, PO3, PO4	Mid Sem Examination (Theory) <table><tr><td colspan="5">Mid Sem Assessment Pattern: 25 Marks</td></tr><tr><td>Sl#</td><td colspan="4">Content</td></tr><tr><td colspan="5">Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)</td></tr><tr><td>1</td><td colspan="3">5 Questions of 1 mark each</td><td>5</td></tr><tr><td colspan="5">Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)</td></tr><tr><td>2</td><td colspan="3">4 Questions of 5 marks each with 2 or 3 sub divisions possibly</td><td>20</td></tr></table>				Mid Sem Assessment Pattern: 25 Marks					Sl#	Content				Part A- 5 Marks (Minimum 1 mark, Max 2 marks, Max 5 Questions)					1	5 Questions of 1 mark each			5	Part B- 20 Marks (Compulsory 4 questions of 5 marks each, there can be max 3 sub divisions)					2	4 Questions of 5 marks each with 2 or 3 sub divisions possibly			20					
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CIE-3 Lab Weightage: 15	CO1-CO4 PO1, PO2,	Lab Write-up and Execution Rubrics: <table><tr><td>Criteria</td><td>Excellent (Full Marks)</td><td>Good (Partial Marks)</td><td>Needs Improvement (No Marks)</td><td>Marks</td></tr><tr><td colspan="5"></td></tr></table>				Criteria	Excellent (Full Marks)	Good (Partial Marks)	Needs Improvement (No Marks)	Marks																														
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	PO3, PO4	<table><tr><td>1. Understanding of Problem Statements</td><td>Demonstrates complete understanding of all problem requirements, constraints, and objectives. (6 marks)</td><td>Shows general understanding with minor misinterpretations or missed constraints. (3–4 marks)</td><td>Poor or incorrect understanding of the problem, missing core elements. (0–2 marks)</td><td>/6</td></tr><tr><td>2. Execution</td><td>Correct and efficient implementation using appropriate logic and data structures. (6 marks)</td><td>Mostly correct with minor errors or less optimal logic. (3–4 marks)</td><td>Incomplete or incorrect execution with major logical flaws. (0–2 marks)</td><td>/6</td></tr><tr><td>3. Result and Documentation</td><td>Results/output are accurate; well-documented with proper explanation, screenshots, formatting. (3 marks)</td><td>Minor documentation issues or minor output mismatches. (1–2 marks)</td><td>Missing/inaccurate output or poorly documented. (0 mark)</td><td>/3</td></tr></table>	1. Understanding of Problem Statements	Demonstrates complete understanding of all problem requirements, constraints, and objectives. (6 marks)	Shows general understanding with minor misinterpretations or missed constraints. (3–4 marks)	Poor or incorrect understanding of the problem, missing core elements. (0–2 marks)	/6	2. Execution	Correct and efficient implementation using appropriate logic and data structures. (6 marks)	Mostly correct with minor errors or less optimal logic. (3–4 marks)	Incomplete or incorrect execution with major logical flaws. (0–2 marks)	/6	3. Result and Documentation	Results/output are accurate; well-documented with proper explanation, screenshots, formatting. (3 marks)	Minor documentation issues or minor output mismatches. (1–2 marks)	Missing/inaccurate output or poorly documented. (0 mark)	/3
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CIE-3 VIVA Weightage: 5	<table><tr><th colspan="8">Viva Voce Rubrics (Max: 4 marks)</th></tr><tr><td>1</td><td>Conceptual Understanding (4 Marks)</td><td>Viva Voce</td><td>Explains thoroughly the core concepts and principles of data structure used in corresponding lab as well as python constructs. (4 M)</td><td>Adequately explains the core concepts and principles of data structure used in corresponding lab as well as python constructs with some understanding (3 M)</td><td>Unable to explain the core concepts and principles of data structure used in corresponding lab as well as python constructs. (0 M)</td><td></td><td></td></tr></table>	Viva Voce Rubrics (Max: 4 marks)								1	Conceptual Understanding (4 Marks)	Viva Voce	Explains thoroughly the core concepts and principles of data structure used in corresponding lab as well as python constructs. (4 M)	Adequately explains the core concepts and principles of data structure used in corresponding lab as well as python constructs with some understanding (3 M)	Unable to explain the core concepts and principles of data structure used in corresponding lab as well as python constructs. (0 M)		
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CIE-3 Record Weightage: 5	CO1-CO4 PO1-PO8	<table><tr><th colspan="4">Lab Record Rubrics:</th></tr><tr><th>Criteria</th><th>Excellent (1 Mark)</th><th>Average (0 Marks)</th><th>Marks</th></tr><tr><td>1. Aim, Apparatus & Theory</td><td>Clearly written aim, complete list of apparatus, and correct explanation of theory.</td><td>Aim/theory unclear or apparatus list incomplete/incorrect.</td><td>/1</td></tr></table>	Lab Record Rubrics:				Criteria	Excellent (1 Mark)	Average (0 Marks)	Marks	1. Aim, Apparatus & Theory	Clearly written aim, complete list of apparatus, and correct explanation of theory.	Aim/theory unclear or apparatus list incomplete/incorrect.	/1			
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1. Aim, Apparatus & Theory	Clearly written aim, complete list of apparatus, and correct explanation of theory.	Aim/theory unclear or apparatus list incomplete/incorrect.	/1														

		2. Circuit Diagram & Truth Table	Correct and neatly drawn circuit diagram with accurate truth table.	Diagram or truth table is missing or incorrect.	/1
		3. Procedure	All steps are clearly written in correct logical order.	Procedure is missing, incomplete, or not logically ordered.	/1
		4. Observation & Result	Observations are accurate and result matches the expected output.	Observations are missing or results are incorrect/incomplete.	/1
		5. Conclusion & Neatness	Clear conclusion based on result, and record is well-organized and neatly presented.	Conclusion is missing or irrelevant, and record is messy.	/1